

Customer Churn Prediction & Retention Investment Strategy

E-commerce · Machine Learning · Business Strategy

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Industry	E-Commerce
Dataset	5,630 customers · 21 features
Models built	Logistic Regression · Decision Tree · Random Forest · XGBoost
Star metric	AUC-ROC (Area Under the ROC Curve)
Winning model	Logistic Regression — AUC-ROC 0.8072
Document type	Stakeholder case study & retention investment brief

1. What This Project Is About

A lot of e-commerce businesses know they have a churn problem. Customers stop buying, stop engaging and eventually disappear but without data, the response tends to be either panic-driven discounting or doing nothing at all. Neither works particularly well.

This project takes a different approach. Working from a dataset of 5,630 customers with 21 behavioural and demographic features, I built a machine learning model that assigns every customer a churn probability score, groups them into risk tiers based on that score and then figures out what you should actually spend on keeping them and whether that spending makes financial sense.

The goal was never just a model that outputs numbers. The goal was something a leadership team could look at and make a decision from. That meant going all the way from raw data to a retention investment plan with a projected ROI.

The four questions this answers

Before starting any analysis, I framed this around four specific business questions because a model that doesn't answer a real question isn't useful:

- Which customers are most likely to leave in the near term?
- What is actually causing them to leave — what are the real drivers?
- Which customers are worth spending money to retain and how much?
- What return should we expect on the retention investment?

Each section of this document works through one of those questions in turn.

2. The Data — What We Had to Work With

The dataset covers 5,630 customers with 21 variables capturing a mix of demographics, behavioural signals, transaction history and customer service interactions. The churn rate in the dataset sits at 24.7% — meaning 1,393 of the 5,630 customers had already churned. That's a reasonably healthy class imbalance for training purposes.

Data quality — what needed fixing

Four columns had missing values, all sitting at exactly 4.99% (281 records each): Tenure, HourSpendOnApp, OrderAmountHikeFromLastYear and DaySinceLastOrder. That's not a coincidence — it almost certainly means the same 281 records were missing across all four, likely newer customers who hadn't been in the system long enough to generate complete data.

All four were imputed with their respective medians: Tenure → 12 months, HourSpendOnApp → 3 hours, DaySinceLastOrder → 2 days, OrderAmountHikeFromLastYear → 14.9%. No records were removed. There were no exact duplicates and no negative values in any numeric column — the dataset was clean apart from those four missing fields.

The features that ended up mattering most

Not every column gets used equally in a model. After training, the features that carried the most predictive weight were:

Feature	What it measures	Importance (Random Forest)
Tenure	How long the customer has been on the platform	0.1805
CLV Proxy	Tenure × Monthly Charges — a lifetime value estimate	0.1390
Satisfaction Score	Customer-reported satisfaction (1–5 scale)	0.1194
Complain	Whether the customer filed a complaint (binary)	0.0866
Cashback Amount	Total cashback received across all orders	0.0550
Monthly Charges	Current monthly spend	0.0529
Engagement Score	Composite of app hours, order count, coupon use	0.0512
Days Since Last Order	How recently the customer placed an order	0.0499

WHAT THE FEATURES TELL US

Tenure and satisfaction score are the two most actionable signals here. A new customer who scores low on satisfaction is carrying two risk factors at once — and they are early enough in their lifecycle that intervention is still possible. Days since last order is the silent indicator: when a customer goes quiet, the clock is already ticking.

3. What the Data Showed Before Any Modelling

Before touching a model, I spent time on the exploratory analysis partly because patterns in raw data often tell you things a model will later confirm and partly because some of the most useful insights for the business team came out of this stage rather than the ML section.

New customers churn at a dramatically higher rate

Customers in their first six months (the 'New' tenure group) churn significantly more than any other group. This isn't surprising in itself — it's a well-known pattern in subscription and repeat-purchase businesses but the scale of the gap matters. Loyal customers with over 36 months on the platform have a churn rate close to single digits, while new customers are multiple times higher.

The practical implication: whatever is happening in the first six months whether it's product experience, customer service, onboarding communications or something else is the primary place to intervene if you want to move the overall churn number.

Complaints are a near-certain churn signal

Customers who filed a complaint had a churn rate roughly three times higher than those who did not. That gap is large enough that 'filed a complaint' should probably be treated as an immediate high-priority flag in any CRM system, regardless of what the model says. The model confirms it, but you do not actually need a machine learning model to act on this one.

The more important question this raises: what is the complaint resolution process currently doing with flagged customers? If the answer is 'sending them to a standard queue,' that process is worth reviewing.

Single customers churn more than married or divorced

Of the marital status segments, single customers showed the highest churn rate. Cash-on-delivery users and mobile phone category shoppers also showed above-average churn. These segments overlap — single customers skew younger and more mobile-first — which suggests there may be a product or experience issue specific to that cohort rather than multiple separate problems.

Low satisfaction scores are almost predictive on their own

The distribution charts for satisfaction score versus churn were stark. Customers scoring 1 or 2 out of 5 churned at rates that were visually obvious even before statistical testing. A customer who tells you they are dissatisfied and you do nothing about it is a customer you are choosing to lose.

THE KEY EDA TAKEAWAY

If you want to act before the model is deployed: pull every customer who has filed a complaint in the last 30 days, cross-reference with satisfaction score 1–2, and call them. That list will have a churn rate somewhere around 50–60%. You do not need machine learning for that intervention.

4. Building and Comparing the Models

I trained four models on the same data split — 80% training (4,504 records), 20% test (1,126 records) with stratified sampling to preserve the 24.7% churn rate in both sets. All models were also validated with 5-fold cross-validation to check they were generalising rather than memorising.

The primary evaluation metric throughout is AUC-ROC (Area Under the Receiver Operating Characteristic Curve). For churn prediction specifically, AUC-ROC is a better metric than accuracy because it measures how well the model separates churners from non-churners across all possible decision thresholds not just at one arbitrary cutoff. An AUC of 0.5 means the model is guessing randomly; 0.8 means it is doing meaningfully better than chance across the full range of predictions.

Model	AUC-ROC	F1 Score	CV AUC (5-fold)	Verdict
Logistic Regression	0.8072	0.5585	0.8178 ± 0.004	✓ Best model — strongest AUC + most stable
XGBoost	0.7817	0.5240	0.7944 ± 0.015	Good — higher variance than LR
Random Forest	0.7783	0.5191	0.7905 ± 0.014	Good — used for feature importance
Decision Tree	0.7516	0.5241	0.7244 ± 0.023	Weakest — high variance, prone to overfitting

Why Logistic Regression won

Logistic Regression came out on top which surprises people who expect tree-based or gradient boosting methods to always outperform simpler approaches. The reason here is probably the relatively modest dataset size (5,630 records) combined with the fact that the churn signal is largely linear: when tenure goes up, churn risk goes down; when satisfaction goes up, churn risk goes down; when complaints go up, churn risk goes up. Linear relationships suit logistic regression well.

The more important signal from the cross-validation results is the variance. Logistic Regression had a standard deviation of ±0.004 across five folds. The Decision Tree had ±0.023. That difference matters — a model with low variance is more likely to perform consistently on data it has never seen, which is what you need for production scoring.

What the best model actually delivers

On the hold-out test set of 1,126 customers:

Metric	Retained Customers	Churned Customers
Precision	89%	45%
Recall	71%	73%
F1 Score	79%	56%
Support	847 customers	279 customers

HOW TO READ THIS TABLE

Precision tells you: when the model predicts churn, how often is it right? For churners, 45% — meaning more than half of predicted churners are false alarms. Recall tells you: of all actual churners, how many did the model catch? 73% — meaning it misses about one in four real churners. For a retention campaign, high recall matters more than high precision. Missing a real churner is expensive. Spending a \$8 email on a false alarm is not.

5. Setting the Right Decision Threshold

Most classification models ship with a default prediction threshold of 0.5 — if the predicted churn probability exceeds 50%, flag as at-risk. For a retention campaign, this is almost always the wrong choice.

The reason comes down to cost asymmetry. Missing an actual churner — someone who leaves because we did not identify them — is significantly more expensive than spending a few pounds on an email sequence for someone who was not going to leave anyway. I modelled this explicitly using the following cost assumptions:

Error Type	What it means	Assumed cost
False Positive	Model flags a non-churner; we spend on retention action unnecessarily	\$5 per customer
False Negative	Model misses a real churner; customer leaves without intervention	\$50 per customer

Running this cost model across every threshold from 0.05 to 0.95 produced a clear minimum at a threshold of 0.21. At that threshold, the combined cost of false positives and false negatives is minimised — which means, in plain terms, that we catch significantly more real churners, at the acceptable cost of also flagging some customers who were not going to leave.

This is the threshold used for all risk scoring in this report. It is deliberately more aggressive than 0.5 and that is the right call given the cost structure of churn.

6. How We Segmented the Customer Base

With a churn probability for every customer, I divided the full base into three risk tiers. The thresholds (70% for high risk, 30% for the boundary between medium and low) were set to create meaningfully different groups rather than arbitrary splits — the actual churn rates within each tier validate the approach.

Risk Tier	Customers	Share	Actual Churn Rate	Avg Monthly Revenue	Annual Revenue at Risk
High Risk ($\geq 70\%$ prob)	1,147	20.4%	60.2%	\$68.58	\$943,924
Medium Risk (30–70%)	2,259	40.1%	25.5%	\$65.88	\$1,785,780
Low Risk ($< 30\%$)	2,224	39.5%	5.6%	\$63.90	—

The validation here is that the model's risk tiers match what actually happened. High Risk customers had a real churn rate of 60.2% — meaning the model correctly identified the customers who were most likely to leave. Low Risk customers churned at only 5.6%.

One thing worth noting: the average monthly revenue is fairly similar across all three tiers (\$63–69). This means you cannot prioritise simply by 'who spends the most.' The model's risk score is the relevant input, not revenue alone.

1,147 High Risk Customers Churn probability $\geq 70\%$	60.2% Actual churn rate (High) Model validated	\$2.73M Annual Revenue at Risk High + Medium combined	0.8072 Model AUC-ROC Logistic Regression
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7. The Retention Plan — What to Do With the Predictions

A churn model is only useful if it drives action. This section is the 'what now' — a structured intervention plan that assigns a specific action and cost to every customer in the at-risk segments, based on their risk tier and revenue profile.

The median monthly charge across the customer base is \$65.47. I used that as the boundary between 'high revenue' and 'lower revenue' within the High Risk tier, because the intervention intensity should match the expected return.

Customer Group	Customers	Intervention	Cost per Customer	Total Budget
High Risk + High Revenue (≥\$65/mo)	617	Personal outreach call + loyalty discount offer	\$20	\$12,340
High Risk + Lower Revenue (<\$65/mo)	530	Automated email sequence + cashback offer	\$8	\$4,240
Medium Risk (all revenue levels)	2,259	Monthly engagement campaign + product recommendations	\$3	\$6,777
Low Risk — New Customers (0–6 months)	210	Structured onboarding programme	\$1	\$210
Low Risk — Established Customers	2,014	Newsletter + personalised product recommendations	\$1	\$2,014

Why these interventions were chosen

The High Risk + High Revenue group gets the most intensive and expensive treatment — personal outreach — because the potential return justifies it. A customer paying \$89/month who you retain for an additional 12 months is worth roughly \$1,068 in prevented churn. Spending \$20 on a phone call and a discount code is a reasonable trade.

The High Risk + Lower Revenue group gets a lower-cost automated intervention. The logic is the same — it is still worth acting — but the economics do not support personalised outreach at that revenue level.

Medium Risk customers get a lighter-touch engagement campaign. They are not imminently leaving, but the data shows they are drifting. Regular engagement at a low cost per customer is the right approach — you are trying to interrupt a trend before it becomes a decision to leave.

New customers in the Low Risk tier get onboarding attention even though their churn probability is low, because we know from the EDA that new customer tenure is the single most predictive churn factor. The intervention there is really prevention rather than rescue.

8. The Financial Case — ROI Analysis

This is the section most data projects skip but it is the one a CFO or operations director actually needs. Everything up to this point is analysis. This is the business case.

The numbers

	High Risk Segment	Medium Risk Segment	Total
Customers targeted	1,147	2,259	3,406
Annual revenue at risk	\$943,924	\$1,785,780	\$2,729,704
Intervention budget	\$16,580	\$6,777	\$23,357
Retention success rate assumed	25%	15%	—
Expected revenue saved	\$235,981	\$267,867	\$503,848
Net benefit	—	—	\$480,491
ROI	—	—	2,057%

\$503K Expected Annual Revenue Saved Conservative estimate	\$23K Total Retention Budget All interventions combined	\$480K Net Benefit Savings minus investment	2,057% Return on Investment \$21.57 saved per \$1 spent
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A NOTE ON THE ROI ASSUMPTIONS

The 25% and 15% retention success rates are deliberately conservative. They assume that one in four High Risk customers and one in six Medium Risk customers will be retained through these interventions. If your retention team is experienced and the campaign execution is strong, actual save rates could be higher. The ROI figure would scale accordingly. What matters is that even at these conservative assumptions, the investment case is overwhelming — you are spending \$23,357 to potentially save \$503,848.

What you are risking by not acting

The total annual revenue at risk across High and Medium Risk customers is \$2,729,704. That number assumes no intervention and a continuation of current churn behaviour. It is not a prediction that all of it will be lost — some of those customers would have stayed regardless — but it represents the exposure you are carrying without a retention programme in place.

The retention programme costs \$23,357 to run. The cost of doing nothing, even if the actual churn rate is half of what the model suggests, is orders of magnitude larger. The risk is asymmetric in favour of acting.

9. Recommendations

These are broken into three-time horizons. The immediate actions do not require the model to be deployed — they follow directly from the exploratory analysis and can start this week. The short-term actions require the model output and some CRM integration. The longer-term roadmap is about making this capability permanent rather than a one-off exercise.

This week — no model required

Action	Why	Who does it
Pull every customer with Complain=1 AND SatisfactionScore ≤ 2 and assign to a retention specialist	This group has a real churn rate close to 50-60%. You do not need a model to identify it or act on it.	Customer Service / CRM Manager
Flag all customers with zero orders in the past 15+ days for a re-engagement email	Days Since Last Order is the fourth strongest predictor. Customers going quiet are often already mentally gone.	Marketing / CRM Team
Brief customer service on complaint resolution SLA — target same-day response for churn-risk indicators	The gap between 'filed a complaint' and 'churn rate' is too large to be a coincidence. Speed of resolution matters.	Operations Manager

Next 1 to 3 months — model-driven actions

Action	Why	Expected Impact
Deploy model scores to CRM; run weekly batch scoring on all active customers	Risk tiers become a live operational input rather than a one-time analysis	Ongoing identification of emerging at-risk customers
Launch personal outreach campaign for the 617 High Risk + High Revenue customers	Highest revenue at risk, highest cost-justified intervention	Direct revenue recovery in 30-60 days
Build automated email sequence for the 530 High Risk + Lower Revenue customers	Low cost, scalable, no manual effort once built	Catch the segment that would otherwise be left without intervention
Review new customer (0–6 month) onboarding — specifically the first 30 days	Tenure is the #1 churn driver. Fixing early-tenure experience moves the whole model's numbers.	Long-term reduction in model's high-risk population

6 to 12 months — making this permanent

- Retrain the model quarterly with fresh data. Churn patterns shift when products, pricing or competitor behaviour change — a model trained on 2024 data will degrade over time.
- Set a monitoring threshold: if AUC-ROC drops below 0.75 on a rolling validation window, trigger an immediate retrain.
- Run A/B tests on the retention interventions — split the High Risk segment between personal outreach and automated email for a quarter, and measure actual save rates. This will give you real retention success rates to replace the conservative 25%/15% assumptions in this ROI model.
- Explore real-time scoring: rather than weekly batch scoring, trigger a churn probability recalculation whenever a customer event occurs (complaint filed, order cancelled, satisfaction score drops). This shortens the window between risk signal and intervention.
- Combine with the customer segmentation model — overlay RFM segments with churn scores to create precision-targeted retention offers based on both purchase behaviour and risk level.

10. Limitations and What to Treat Carefully

No analysis is complete without being honest about what it does not do well. These are the limitations worth keeping in mind when acting on this work.

The model does not know why customers are unhappy

Satisfaction score is the third strongest predictor of churn, but the model treats it as a number — it does not know whether that low score reflects a product quality issue, a delivery problem, a customer service interaction or something about the website experience. The model tells you who is at risk; it does not tell you what to say to them when you call. That requires qualitative follow-up.

The ROI projections are estimates, not guarantees

The 25% and 15% retention success rates used in the financial model are assumptions, not measurements from this dataset. They are conservative by design — the actual rates will depend heavily on how well the interventions are executed, the quality of the outreach and what you are offering. Track actual save rates from the first campaign cycle and update the model assumptions accordingly.

The model will degrade without maintenance

Customer behaviour changes. If the business changes its pricing, launches new product categories or shifts its marketing mix, the relationships between features and churn may shift. A quarterly retrain with fresh data and a monitored AUC metric will catch this before it becomes a problem.

False positives will frustrate some customers

At the 0.21 decision threshold, the model flags a meaningful number of customers as at-risk who were not actually going to leave. These customers will receive retention outreach — discounts, calls, offers — that they did not need. At small scale this is fine. At large scale, there is a risk of training customers to expect retention offers before they are at risk, which can actually reduce loyalty rather than build it. Monitor this effect as the programme scales.

Appendix — Technical Summary

Model performance in full

Model	AUC-ROC	F1 Score	CV AUC Mean	CV AUC Std Dev
Logistic Regression	0.8072	0.5585	0.8178	±0.0040
XGBoost	0.7817	0.5240	0.7944	±0.0149
Random Forest	0.7783	0.5191	0.7905	±0.0137
Decision Tree	0.7516	0.5241	0.7244	±0.0230

Engineered features created

Feature	Formula / Logic	Purpose
CLV_Proxy	Tenure × MonthlyCharges	Proxy for customer lifetime value without a full CLV model
TenureGroup	Bucketed: New (0–6m), Growing (7–18m), Established (19–36m), Loyal (36m+)	Allows categorical analysis of tenure effects
RevenueSegment	Bucketed MonthlyCharges: Low (<\$40), Medium (\$40–75), High (>\$75)	Revenue-based segmentation for intervention design
EngagementScore	HoursOnApp×2 + OrderCount×1.5 + CouponUsed×0.5 – DaySinceLastOrder×0.3	Composite engagement signal across multiple behaviours
HighRiskFlag	1 if SatisfactionScore≤2 OR Complain=1 OR DaySinceLastOrder≥10	Rule-based pre-screen — useful for CRM teams without model access

Visualisations produced (13 charts)

- 1_churn_distribution.png — overall churn rate breakdown
- 2_churn_by_category.png — churn rate by marital status, payment mode, order category, city tier, device, gender
- 3_numeric_vs_churn.png — tenure, satisfaction, charges, recency distributions by churn status
- 4_complain_vs_churn.png — complaint impact on churn rate
- 5_engineered_features.png — churn rate by tenure group, revenue segment and high risk flag
- 6_auc_roc.png — ROC curves for all four models
- 7_confusion_matrices.png — confusion matrices for all four models
- 8_feature_importance.png — Random Forest and XGBoost feature importance
- 9_threshold_optimisation.png — F1 and business cost curves across decision thresholds
- 10_risk_segmentation.png — risk tier distribution, churn probability histogram, revenue by tier
- 11_intervention_plan.png — customers and budget per intervention type
- 12_roi_analysis.png — revenue at risk vs expected savings, investment waterfall, ROI gauge
- 13_executive_dashboard.png — single-page summary for leadership briefing

Files exported

- data/processed/ecommerce_churn_clean.csv — cleaned dataset with engineered features
- data/processed/ecommerce_churn_powerbi.csv — full scored dataset including ChurnProbability, RiskTier, InterventionType, InterventionCost
- data/processed/model_performance_summary.csv — all four models' metrics
- models/best_churn_model.pkl — trained Logistic Regression model with scaler and label encoders

